Certificate

By completing the certificate in Supply Chain Management, students will learn to:

- Design and manage effective supply chains
- Improve the operational efficiencies of a supply chain
- Be able to interpret the implications of supply chain initiatives in terms of key business performance and analytical metrics
- Apply supply chain analytical tools and decision-making for supply chain analysis.
- Recognize and evaluate supply chain vulnerabilities, mitigate supply chain risk, and build resilient domestic and global supply chains
- Evaluate and manage the sustainability of a supply chain

A BS or BA degree in any discipline with a GPA of > 3.0 is required for this program.

Program Requirements

The graduate certificate in Supply Chain Management requires completion of three required courses and one elective for a total of twelve credits.

Three required courses (9 credits)

Item #	Title	Credits
OIE 501	Operations Management	3
OIE 542	Risk Management and Decision Analysis	3
OIE 544	Supply Chain Analysis and Design	3

One elective (3 credits)

Choose one of the following elective courses:

Item #	Title	Credits
OIE 549	Sustainable Supply Chain and Operations Management	3
OIE 553	Global Purchasing and Logistics	3
OIE 559	Advanced Prescriptive Analytics: From Data to Impact	3

Customized Certificate

Degree Type

Certificate

Custom Certificates

Students may customize their own four-course, 12-credit graduate business certificate. Those interested in a customized certificate may provide a rationale for certificate makeup to the executive director of business programs at business-prog-office@wpi.edu.

Master of Business Administration (MBA)

Degree Type

M.B.A.

WPI's MBA program is aimed at STEM professionals seeking the skills to strategically lead organizations. The curriculum features a 5-course core covering the business disciplines in the context of tech-driven environments; an integrative STEM course; a 3-course tech management specialization; 2 customizable electives; and a capstone course.

MBA Options

The MBA is available fully online via a mix of asynchronous and optional synchronous deliveries. Some MBA courses are also available on campus. The full MBA cannot be completed entirely on campus.

Required Courses (5 courses)

Item #	Title	Credits
FIN 500	Financial Management	3
MIS 584	Business Intelligence	3
MKT 500	Marketing Strategy	3
OBC 506	Leadership	3
OIE 501	Operations Management	3

MBA Specialty (3 courses)

Item #	Title	Credits
	Specialty in Advanced Business Analytics Methods	
	Specialty in Advanced Operations Analytics	
	Specialty in Applied Analytics	
	Specialty in Artificial Intelligence in Business	
	Specialty in Brand, Products, and Consumers	
	Specialty in Business Analytics	
	Specialty in Data Analytics	
	Specialty in Digital Transformation	
	Specialty in Entrepreneurship	
	Specialty in Information Systems Design	
	Specialty in Information Technology	
	Specialty in Information Technology User Experience	
	Specialty in Marketing Analytics	
	Specialty in Operational Excellence	
	Specialty in Operations Analytics	
	Specialty in Organizing and Managing Innovation	
	Specialty in Product Management	
	Specialty in Project Management	
	Specialty in Supply Chain Management	
	Specialty in System Design for IUX	

Elective Courses (2 courses)

Must be Business School courses

Integrative STEM Course (1 course)

Item #	Title	Credits
BUS 590	Strategic Management	3

Capstone (1 course)

Item #	Title	Credits
BUS 599	Capstone Project	3

Specialty in Advanced Business Analytics Methods

Item #	Title	Credits
MIS 587	Business Applications in Machine Learning	3
MKT 562	Marketing Research	3
OIE 559	Advanced Prescriptive Analytics: From Data to Impact	3

Specialty in Advanced Operations Analytics

Select three courses from the following:

(At least two of the three courses must be OIE courses.)

Item #	Title	Credits
OIE 542	Risk Management and Decision Analysis	3
OIE 548	Performance Analytics	3
OIE 559	Advanced Prescriptive Analytics: From Data to Impact	3
MIS 587	Business Applications in Machine Learning	3
MKT 562	Marketing Research	3

Specialty in Applied Analytics

Select three of the following courses:

Item #	Title	Credits
MIS 502	Data Management for Analytics	3
MIS 584	Business Intelligence	3
MIS 587	Business Applications in Machine Learning	3
OIE 559	Advanced Prescriptive Analytics: From Data to Impact	3
DS 501	Introduction to Data Science	3
DS 502/MA 543	Statistical Methods for Machine Learning	3

*MA 511 can be substituted for DS 502/MA 543

Specialty in Artificial Intelligence in Business

Item #	Title	Credits
MIS 587	Business Applications in Machine Learning	3

Select one of the following courses:

Item #	Title	Credits
MIS 520	Artificial Intelligence and its Ethical Applications in Business	3
MIS 581	Policy and Strategy for Information Technology, Artificial Intelligence, and Analytics	3

Select one of the following courses:

Item #	Title	Credits
MKT 562	Marketing Research	3
MKT 568	Marketing Analytics	3
OIE 552	Modeling and Optimizing Processes	3

Specialty in Brand, Products, and Consumers

Select three of the following courses:

Item #	Title	Credits
MKT 500	Marketing Strategy	3
MKT 561	Consumer Behavior and Analytics	3
MKT 562	Marketing Research	3
MKT 565	Digital and Social Media Marketing	3
MKT 568	Marketing Analytics	3
MKT 569	Product Management	3

Specialty in Business Analytics

Item #	Title	Credits
MIS 502	Data Management for Analytics	3
MIS 584	Business Intelligence	3
OIE 552	Modeling and Optimizing Processes	3

Specialty in Data Analytics

Item #	Title	Credits
MIS 587	Business Applications in Machine Learning	3
MKT 568	Marketing Analytics	3

Select one of the following courses:

Item #	Title	Credits
MKT 562	Marketing Research	3
MKT 565	Digital and Social Media Marketing	3

Specialty in Digital Transformation

Item #	Title	Credits
MIS 500	Innovating with Information Systems	3

Select one of the following:

Item #	Title	Credits
OIE 501	Operations Management	3
OIE 544	Supply Chain Analysis and Design	3
OIE 558	Designing and Managing Lean Six Sigma Processes	3

Select one of the following:

Item #	Title	Credits
MIS 576	Project Management	3
OBC 533	Negotiations	3
OIE 558	Designing and Managing Lean Six Sigma Processes	3
OBC 537	Leading Change	3

Specialty in Entrepreneurship

Item #	Title	Credits
ETR 500	Entrepreneurship and Innovation	3

Choose two of the following courses:

Item #	Title	Credits
BUS 500	Business Law, Ethics and Social Responsibility	3
BUS 546	Managing Technological Innovation	3
ETR 596	Selling and Sales	3
ETR 593	Technology Commercialization	3

Specialty in Information Systems Design

Item #	Title	Credits
MIS 571	Database Applications Design and Development	3
MIS 582	Information Security Management	3
MIS 585	User Experience Design	3

Specialty in Information Technology

Item #	Title	Credits
MIS 502	Data Management for Analytics	3
MIS 581	Policy and Strategy for Information Technology, Artificial Intelligence, and Analytics	3
MIS 584	Business Intelligence	3

Specialty in Information Technology User Experience

Item #	Title	Credits
MIS 583	User Experience Applications	3
MIS 586	User Experience Methods	3
MKT 561	Consumer Behavior and Analytics	3

Specialty in Marketing Analytics

Item #	Title	Credits
MKT 568	Marketing Analytics	3

Select two of the following courses:

Item #	Title	Credits
MKT 500	Marketing Strategy	3
MKT 561	Consumer Behavior and Analytics	3
MKT 565	Digital and Social Media Marketing	3

Specialty in Operational Excellence

Choose one course from:

Item #	Title	Credits
OIE 554	Global Operations Strategy	3
OIE 558	Designing and Managing Lean Six Sigma Processes	3

Choose two courses from:

Item #	Title	Credits
FIN 500	Financial Management	3
MIS 576	Project Management	3
OBC 537	Leading Change	3
OIE 554	Global Operations Strategy	3
OIE 558	Designing and Managing Lean Six Sigma Processes	3

Specialty in Operations Analytics

Select three of the following courses:

Item #	Title	Credits
OIE 501	Operations Management	3
OIE 542	Risk Management and Decision Analysis	3
OIE 544	Supply Chain Analysis and Design	3
OIE 548	Performance Analytics	3
OIE 558	Designing and Managing Lean Six Sigma Processes	3

Specialty in Organizing and Managing Innovation

Select three of the following courses:

Item #	Title	Credits
BUS 546	Managing Technological Innovation	3
ETR 500	Entrepreneurship and Innovation	3
ETR 593	Technology Commercialization	3
FIN 500	Financial Management	3
MIS 576	Project Management	3
OBC 505	Teaming and Organizing for Innovation	3
OBC 533	Negotiations	3
OBC 537	Leading Change	3

Specialty in Product Management

Item #	Title	Credits
MKT 569	Product Management	3

Choose two of the following courses:

Item #	Title	Credits
BUS 546	Managing Technological Innovation	3
ETR 593	Technology Commercialization	3
MIS 576	Project Management	3
MIS 583	User Experience Applications	3
MIS 585	User Experience Design	3
MKT 565	Digital and Social Media Marketing	3
OBC 535	Managing Creativity in Knowledge Intensive Organizations	3
OBC 533	Negotiations	3
OBC 505	Teaming and Organizing for Innovation	3

Specialty in Project Management

Item #	Title	Credits
MIS 576	Project Management	3

Select two of the following:

Item #	Title	Credits
BUS 546	Managing Technological Innovation	3
OBC 537	Leading Change	3
OBC 535	Managing Creativity in Knowledge Intensive Organizations	3
OBC 533	Negotiations	3
OBC 505	Teaming and Organizing for Innovation	3

Specialty in Supply Chain Management

Choose one course from:

Item #	Title	Credits
OIE 553	Global Purchasing and Logistics	3
OIE 549	Sustainable Supply Chain and Operations Management	3

Choose two courses from:

Item #	Title	Credits
MKT 561	Consumer Behavior and Analytics	3
OBC 533	Negotiations	3
OIE 553	Global Purchasing and Logistics	3
OIE 548	Performance Analytics	3
OIE 549	Sustainable Supply Chain and Operations Management	3

Specialty in System Design for IUX

Select three of the following courses:

Item #	Title	Credits
MIS 500	Innovating with Information Systems	3
MIS 571	Database Applications Design and Development	3
MIS 582	Information Security Management	3
IMGD 5000	IMGD Studio	3
IMGD 5300	Design of Interactive Experiences	3
RBE 526/CS 526	Human-Robot Interaction	3
RBE 595	Special Topics	2-3
WR 593	Special Topics in Writing and Communication	3

Master of Business Administration (MBA), Analytics

Degree Type

M.B.A.

Program Goals and Objectives

The STEM-certified Analytics MBA is designed for graduate business students early in their career who are interested in developing analytics capabilities and expect to move into starting managerial positions. The program will be STEM designated by the USCIS, thus opening a path to business analytics and managerial careers for international students.

- Equips students with analytic methods, tools, techniques needed for entry-level jobs at the intersection of analytics, strategy, business, and management.
- Enables students to interpret analytic results, draw insights to make strategic business decisions, and recommend additional analysis that could aid data-driven strategy formulation.
- Prepares students for the management jobs they will encounter early in their career, including project management and product management.

Faculty Contacts:

Professors Purvi Shah, Bengisu Tulu, Jim Ryan, Renata Konrad

Core Courses (Four Courses)

Item #	Title	Credits
MIS 502	Data Management for Analytics	3
FIN 500	Financial Management	3
MKT 500	Marketing Strategy	3
OIE 501	Operations Management	3

Specialties (18 Credits, Select Two Specialties of Three Courses Each)

Item #	Title	Credits
	Applied Business Analytics	9
	Data-driven Strategic Management	9
	Marketing Analytics	9
	Operations Analytics	9

Required Capstone Courses

Item #	Title	Credits
OBC 505	Teaming and Organizing for Innovation	3
BUS 594	Data-driven Business Strategy (Capstone)	3

Applied Business Analytics

9

Select any three of the following:

Item #	Title	Credits
FIN 540	Financial Analytics	3
MIS 584	Business Intelligence	3
MIS 587	Business Applications in Machine Learning	3
MKT 568	Marketing Analytics	3
OIE 552	Modeling and Optimizing Processes	3

Data-driven Strategic Management

9

Select any three of the following:

Item #	Title	Credits
BUS 546	Managing Technological Innovation	3
ETR 593	Technology Commercialization	3
MIS 576	Project Management	3
MKT 569	Product Management	3

Marketing Analytics

9

Item #	Title	Credits
MKT 568	Marketing Analytics	3

Select two of the following:

Item #	Title	Credits
MKT 561	Consumer Behavior and Analytics	3
MKT 562	Marketing Research	3
MKT 565	Digital and Social Media Marketing	3

Operations Analytics

9

Select any three of the following:

Item #	Title	Credits
OIE 501	Operations Management	3
OIE 542	Risk Management and Decision Analysis	3
OIE 544	Supply Chain Analysis and Design	3
OIE 548	Performance Analytics	3
OIE 558	Designing and Managing Lean Six Sigma Processes	3
OIE 549	Sustainable Supply Chain and Operations Management	3

M.S. in Business Analytics (MSBA)

Degree Type

Master of Science

The demand for knowledgeable professionals who understand both data analytics and business needs continues to increase. The MSBA program provides a solid foundation in business analytics, with applications in a variety of business areas, plus a real-world, team-based project, which ensures that students are prepared to be successful in a data-driven, technological business environment.

The MSBA is available both on-campus and online.

Required Core Courses (3 courses)

Item #	Title	Credits
MIS 502	Data Management for Analytics	3
MIS 584	Business Intelligence	3
OIE 552	Modeling and Optimizing Processes	3

Two Specialties (6 courses)

MSBA students must complete two three-course specialties as detailed below.

Item #	Title	Credits
	Specialty in Advanced Business Analytics Methods	
	Specialty in Artificial Intelligence in Business	
	Specialty in Marketing Analytics	
	Specialty in Operations Analytics	

Capstone Project Experience (2 courses)

Item #	Title	Credits
OBC 505	Teaming and Organizing for Innovation	3
BUS 596	Master of Science Capstone Project	3

Additional Recommendations:

On-campus, international students are encouraged to complete a zero credit internship to ensure their readiness for employment in the U.S. Internships do not count towards the minimum degree requirements.

Specialty in Advanced Business Analytics Methods

Item #	Title	Credits
MIS 587	Business Applications in Machine Learning	3
MKT 562	Marketing Research	3
OIE 559	Advanced Prescriptive Analytics: From Data to Impact	3

Specialty in Artificial Intelligence in Business

Item #	Title	Credits
MIS 587	Business Applications in Machine Learning	3

Select one of the following courses:

Item #	Title	Credits
MIS 520	Artificial Intelligence and its Ethical Applications in Business	3
MIS 581	Policy and Strategy for Information Technology, Artificial Intelligence, and Analytics	3

Select one of the following courses:

Item #	Title	Credits
MKT 562	Marketing Research	3
MKT 568	Marketing Analytics	3
OIE 552	Modeling and Optimizing Processes	3

Specialty in Marketing Analytics

Item #	Title	Credits
MKT 568	Marketing Analytics	3

Select two of the following courses:

Item #	Title	Credits
MKT 500	Marketing Strategy	3
MKT 561	Consumer Behavior and Analytics	3
MKT 565	Digital and Social Media Marketing	3

Specialty in Operations Analytics

Select three of the following courses:

Item #	Title	Credits
OIE 501	Operations Management	3
OIE 542	Risk Management and Decision Analysis	3
OIE 544	Supply Chain Analysis and Design	3
OIE 548	Performance Analytics	3
OIE 558	Designing and Managing Lean Six Sigma Processes	3

M.S. in Financial Technology (MSFinTech)

Degree Type
Master of Science

There is a growing demand for FinTech talent that can operate in a world of unstructured financial data. This new FinTech talent needs to be familiar with unstructured data, predictive analytics, and artificial intelligence (A.I.) insights to drive new financial thinking and processes that deliver innovative financial products and strategies for long-term productivity improvements and economic growth. The FinTech program prepares students to develop key competencies in predictive analytics and programming applications for quantitative risk management, financial forecasting, corporate innovation, and financial modeling. Graduates from the M.S. FinTech program will possess the financial and applied technical skills to transform business and business practices through innovations that will undoubtedly shape the nature and boundaries of firms in the coming years. These skills will be invaluable to both established firms and startups as we move rapidly to a digitized and data-intensive world.

The M.S. in FinTech is available on-campus.

Required Core Courses (3 courses)

Item #	Title	Credits
FIN 510/MIS 510	Business Application of Blockchain Technology	3
FIN 540	Financial Analytics	3

Two Specialties (6 courses)

MSFinTech students must complete two three-course specialties, selected from the following specialties:

Item #	Title	Credits
	Specialty in Advanced Financial Mathematics	9
	Specialty in FinTech Analytics	9
	Specialty in FinTech Development	9

Capstone Experience (2 courses)

Item #	Title	Credits
OBC 505	Teaming and Organizing for Innovation	3
BUS 596	Master of Science Capstone Project	3

Additional Requirements

On-campus, international students are encouraged to complete up to three additional credits of internship to ensure their readiness for employment in the U.S. Additionally, applicants to the MS FinTech program must have prior college-level statistics and finance/accounting preparation.

Specialty in Advanced Financial Mathematics

9

One of the following:

Item #	Title	Credits
MIS 587	Business Applications in Machine Learning	3

Two of the following:

Item #	Title	Credits
MA 571	Financial Mathematics I	3
MA 572	Financial Mathematics II	3
MA 573	Computational Methods of Financial Mathematics	3
MA 574	Portfolio Valuation and Risk Management	3
MA 575	Market and Credit Risk Models and Management	3
MA 590	Special Topics	3

Specialty in FinTech Analytics

9

One of the following:

Item #	Title	Credits
MA 520	Fourier Transforms and Distributions	3
MIS 587	Business Applications in Machine Learning	3

Two of the following:

Item #	Title	Credits
DS 502/MA 543	Statistical Methods for Machine Learning	3
MIS 502	Data Management for Analytics	3
DS 503/CS 585	Big Data Management	3
CS/DS 541	Deep Learning	3
OIE 559	Advanced Prescriptive Analytics: From Data to Impact	3
MIS 571	Database Applications Design and Development	3
MIS 584	Business Intelligence	3

Specialty in FinTech Development

9
One of the following courses:

Item #	Title	Credits
MIS 587	Business Applications in Machine Learning	3

Two of the following courses:

For those who have taken CS 5007, CS 5008, or equivalent, we recommend taking CS 528.

Item #	Title	Credits
CS 5007	Introduction to Applications of Computer Science with Data Structures and Algorithms	3
CS 5008	Introduction to Systems and Network Programming	3
CS 5084	Introduction to Algorithms: Design and Analysis	3
CS 513	Computer Networks	3
CS 528	Mobile and Ubiquitous Computing	3
CS 573	Data Visualization	3
CS 578/ECE 578	Cryptography and Data Security	3
MIS 583	User Experience Applications	3

M.S. in Information Technology (MSIT)

Degree Type
Master of Science

The demand for information technology professionals who understand technology, analytics, and business needs continues to increase. The MSIT program guarantees a solid foundation in information technology and business analytics, with cutting-edge concentrations, and how these technologies and analytics tools can best be used in a variety of business applications. The program covers principles of business, people and technology that are critical to success in technology-driven environments.

The MSIT is available both on-campus and online.

Required Core Courses (3 courses)

Item #	Title	Credits
MIS 502	Data Management for Analytics	3
MIS 581	Policy and Strategy for Information Technology, Artificial Intelligence, and Analytics	3
MIS 584	Business Intelligence	3

Two Specialties (6 courses)

MSIT students must complete two three-course specialties, selected from the following specialties:

Item #	Title	Credits
	Specialty in Artificial Intelligence in Business	
	Specialty in Information Systems Design	
	Specialty in Data Analytics	
	Specialty in Digital Transformation	
	Specialty in Information Technology User Experience	

Capstone Project Experience (2 courses)

Item #	Title	Credits
OBC 505	Teaming and Organizing for Innovation	3
MIS 573	System Design and Development	3

Additional Recommendations:

On-campus, international students are encouraged to complete a zero credit internship to ensure their readiness for employment in the U.S. Internships do not count towards the minimum degree requirements.

Specialty in Artificial Intelligence in Business

Item #	Title	Credits
MIS 587	Business Applications in Machine Learning	3

Select one of the following courses:

Item #	Title	Credits
MIS 520	Artificial Intelligence and its Ethical Applications in Business	3
MIS 581	Policy and Strategy for Information Technology, Artificial Intelligence, and Analytics	3

Select one of the following courses:

Item #	Title	Credits
MKT 562	Marketing Research	3
MKT 568	Marketing Analytics	3
OIE 552	Modeling and Optimizing Processes	3

Specialty in Information Systems Design

Item #	Title	Credits
MIS 571	Database Applications Design and Development	3
MIS 582	Information Security Management	3
MIS 585	User Experience Design	3

Specialty in Data Analytics

Item #	Title	Credits
MIS 587	Business Applications in Machine Learning	3
MKT 568	Marketing Analytics	3

Select one of the following courses:

Item #	Title	Credits
MKT 562	Marketing Research	3
MKT 565	Digital and Social Media Marketing	3

Specialty in Digital Transformation

Item #	Title	Credits
MIS 500	Innovating with Information Systems	3

Select one of the following:

Item #	Title	Credits
OIE 501	Operations Management	3
OIE 544	Supply Chain Analysis and Design	3
OIE 558	Designing and Managing Lean Six Sigma Processes	3

Select one of the following:

Item #	Title	Credits
MIS 576	Project Management	3
OBC 533	Negotiations	3
OIE 558	Designing and Managing Lean Six Sigma Processes	3
OBC 537	Leading Change	3

Specialty in Information Technology User Experience

Item #	Title	Credits
MIS 583	User Experience Applications	3
MIS 586	User Experience Methods	3
MKT 561	Consumer Behavior and Analytics	3

M.S. in Innovation with User Experience (MSIUX)

Degree Type
Master of Science

Rapid advances in science and engineering allow companies to develop increasingly sophisticated IT products. As the IT industry matures, competition is increasingly shifting toward providing outstanding user experiences (UX). Innovation with UX is becoming essential in developing IT products and services that can maintain competitive advantage in the marketplace. The Business School has world-class expertise and resources in UX and is ideally positioned to prepare students as UX professionals and set them on a path to take on leaderships positions such as chief experience officers (CXO).

The MSIUX is delivered entirely online. Some courses are also available on campus.

Required Core Courses (3 Courses)

Item #	Title	Credits
MIS 583	User Experience Applications	3
MIS 585	User Experience Design	3
MIS 586	User Experience Methods	3

Two Specialties (6 courses)

MSIUX students must complete two, three-course specialties, selected from the following:

Item #	Title	Credits
	Specialty in Brand, Products, and Consumers	
	Specialty in System Design for IUX	
	Specialty in Applied Analytics	
	Specialty in Organizing and Managing Innovation	

Capstone Project Experience (2 courses)

Item #	Title	Credits
OBC 505	Teaming and Organizing for Innovation	3
MIS 573	System Design and Development	3

Additional Recommendations:

On-campus, international students are encouraged to complete a zero credit internship to ensure their readiness for employment in the U.S. Internships do not count towards the minimum degree requirements.

Specialty in Brand, Products, and Consumers

Select three of the following courses:

Item #	Title	Credits
MKT 500	Marketing Strategy	3
MKT 561	Consumer Behavior and Analytics	3
MKT 562	Marketing Research	3
MKT 565	Digital and Social Media Marketing	3
MKT 568	Marketing Analytics	3
MKT 569	Product Management	3

Specialty in System Design for IUX

Select three of the following courses:

Item #	Title	Credits
MIS 500	Innovating with Information Systems	3
MIS 571	Database Applications Design and Development	3
MIS 582	Information Security Management	3
IMGD 5000	IMGD Studio	3
IMGD 5300	Design of Interactive Experiences	3
RBE 526/CS 526	Human-Robot Interaction	3
RBE 595	Special Topics	2-3
WR 593	Special Topics in Writing and Communication	3

Specialty in Applied Analytics

Select three of the following courses:

Item #	Title	Credits
MIS 502	Data Management for Analytics	3
MIS 584	Business Intelligence	3
MIS 587	Business Applications in Machine Learning	3
OIE 559	Advanced Prescriptive Analytics: From Data to Impact	3
DS 501	Introduction to Data Science	3
DS 502/MA 543	Statistical Methods for Machine Learning	3

*MA 511 can be substituted for DS 502/MA 543

Specialty in Organizing and Managing Innovation

Select three of the following courses:

Item #	Title	Credits
BUS 546	Managing Technological Innovation	3
ETR 500	Entrepreneurship and Innovation	3
ETR 593	Technology Commercialization	3
FIN 500	Financial Management	3
MIS 576	Project Management	3
OBC 505	Teaming and Organizing for Innovation	3
OBC 533	Negotiations	3
OBC 537	Leading Change	3

M.S. in Operations and Supply Chain Analytics (MSOSCA)

Degree Type
Master of Science

The MSOSCA prepares students with the leadership skills to facilitate operational excellence. Students explore a full range of operations topics, including risk management, lean process design, modeling & optimization, and global strategy. By learning to identify operational efficiencies in processes and supply chains, students will be able to conduct demand forecasting, evaluate vulnerabilities and risk, and more. Students learn the skills and analytical techniques necessary to design and manage effective processes and supply chains.

The MSOSCA is available both on-campus and online.

Required Core Courses (3 courses)

Item #	Title	Credits
OIE 501	Operations Management	3
OIE 544	Supply Chain Analysis and Design	3
OIE 552	Modeling and Optimizing Processes	3

Two Specialties (6 courses)

MSOSCA students must complete two, three-course specialties, selected from the following:

Item #	Title	Credits
	Specialty in Supply Chain Management	
	Specialty in Advanced Operations Analytics	
	Specialty in Operational Excellence	

Capstone Project Experience (2 courses)

Item #	Title	Credits
OBC 505	Teaming and Organizing for Innovation	3
OIE 597	Operations and Supply Chain Consulting Project	3

Additional Recommendations

On-campus, international students are encouraged to complete a zero credit internship to ensure their readiness for employment in the U.S. Internships do not count towards the minimum degree requirements.

Specialty in Supply Chain Management

Choose one course from:

Item #	Title	Credits
OIE 553	Global Purchasing and Logistics	3
OIE 549	Sustainable Supply Chain and Operations Management	3

Choose two courses from:

Item #	Title	Credits
MKT 561	Consumer Behavior and Analytics	3
OBC 533	Negotiations	3
OIE 553	Global Purchasing and Logistics	3
OIE 548	Performance Analytics	3
OIE 549	Sustainable Supply Chain and Operations Management	3

Specialty in Advanced Operations Analytics

Select three courses from the following:

(At least two of the three courses must be OIE courses.)

Item #	Title	Credits
OIE 542	Risk Management and Decision Analysis	3
OIE 548	Performance Analytics	3
OIE 559	Advanced Prescriptive Analytics: From Data to Impact	3
MIS 587	Business Applications in Machine Learning	3
MKT 562	Marketing Research	3

Specialty in Operational Excellence

Choose one course from:

Item #	Title	Credits
OIE 554	Global Operations Strategy	3
OIE 558	Designing and Managing Lean Six Sigma Processes	3

Choose two courses from:

Item #	Title	Credits
FIN 500	Financial Management	3
MIS 576	Project Management	3
OBC 537	Leading Change	3
OIE 554	Global Operations Strategy	3
OIE 558	Designing and Managing Lean Six Sigma Processes	3

Ph.D. Business Administration

Degree Type
Ph.D.

The PhD in Business Administration at WPI has both traditional and executive options.

Degree Requirements

The course of study leading to the Ph.D. degree in Business Administration requires the completion of 90 credits beyond the bachelor's degree, or 60 credits beyond the master's degree. For students proceeding from master's to Ph.D. degree, the 60 credits should be distributed as follows:

Ph.D. (60 credits)

Item #	Title	Credits
	Courses in Business Administration	12
	18 credits in additional coursework, dissertation research, and supplemental research	18

For students proceeding directly from B.S. degree to Ph.D. degree, the 90 credits should be distributed as follows:

Executive PhD Option

The Executive PhD is designed for seasoned business professionals who are ready to apply their expertise in new and fulfilling ways. Whether you're looking to elevate what you can do in your current role or you want to branch out into consulting, teaching, or conducting research, the Executive PhD program will give you added credibility to become a more impactful thought leader and address broad challenges facing your industry.

The Executive PhD requires a Master's degree, and is thus a PhD 60 program. It also requires at least 15 years of work experience.

The Executive PhD dissertation integrates your industry experience and passion with expert faculty guidance to address key problems in your field of interest.